

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Intelligent Emergency Response and Traffic Network Optimization for Smart Cities

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>AlgoNexus</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Emergency response teams like ambulances, fire trucks, and police units require rapid access to their destinations. In many cities, however, there are many factors which make accessing these locations difficult. For example, many of the routes used by emergency response teams will have multiple stop signs and/or traffic signals. Even though each stop sign/traffic light will be delayed only slightly, when you add together those delays for three to five signals, they become significant. Delays during an emergency situation may result in serious consequences. The reason we chose to model this problem is to apply data structures and algorithmic techniques to a very tangible and relevant issue. We plan to develop our representation of the city's street grid as a graph (intersections = nodes; streets = edges). Using this graph representation, we will be able to determine what is likely to be the most efficient route for an emergency unit to travel from a starting point to a destination. After determining the best route, we will identify the key stoplights along that route and provide them with priority status. At this time, we are developing this program as a software simulation tool. Therefore, we will create some sample representations of the roadways, simulate various amounts of traffic flow on the roadway(s), and evaluate the performance of both the routing technique and the prioritization of the stoplight. We do not expect to develop any hardware such as traffic lights, sensors, etc., so that we can complete the development of our prototype.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Currently, most emergency vehicles are forced to follow the same streets and utilize the same traffic signals as all other road users. There exist some timing-based signals (i.e., "red lights") and adaptive signals that will slightly modify their cycle in response to present traffic conditions. Additionally, there exists specialized equipment designed specifically for emergency services vehicles (e.g., devices that provide a prioritized approach through intersections or tools that recommend a particular path). These work independently of one another. The issue lies in that, generally speaking, they do not communicate with one another. Therefore, a vehicle such as an ambulance or fire engine may have taken a favorable route but continue to be stopped at a red light due to a lack of communication between the traffic signal and the approaching emergency vehicle. There exist several newer systems designed to address this deficiency; utilizing real-time traffic data and allowing vehicles and infrastructure to communicate directly, and/or utilizing signals that can respond immediately to changing traffic conditions. This is the gap that our research focuses upon. As opposed to creating two distinct models that treat routing and providing priority at signals as two individual problems, our proposed model integrates both into one model. First, the model selects a route for the vehicle, and secondly determines which specific signals along that selected route would be triggered during the course of travel—thereby only providing priority to those signals along that route and none others.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

Our main goal here is a work in prototype —software that can actually help an emergency vehicle get where it's going without sitting around at lights it didn't need to. Specifically, we want to test whether handling route selection and signal priority together actually performs better than handling them separately, which seems to be how most current systems work. We'll start by looking in to what's already out there existing work on emergency vehicle routing, signal control, and priority systems. From there, we'll build a small road network to run our simulation on, where each intersection is a node and each road connecting them is an edge. Once the graph is in place, the next step is route-finding: given a source and a destination, the program should be able to work out a reasonable path between them. After that, we'll layer in the priority mechanism—this is the part that identifies which signals fall along the chosen route and adjusts them for the emergency vehicle. We'll test this across a few different road layouts and traffic conditions to see how it holds up. Finally, we'll compare the "normal" case against the optimize done, mainly looking at three things: total route distance, estimated travel time, and how long the vehicle waits at intersections. Assuming the core prototype works as expected, there's room to extend it later—real-time traffic feeds, multiple emergency vehicles at once, or letting the route adapt dynamically as conditions change.

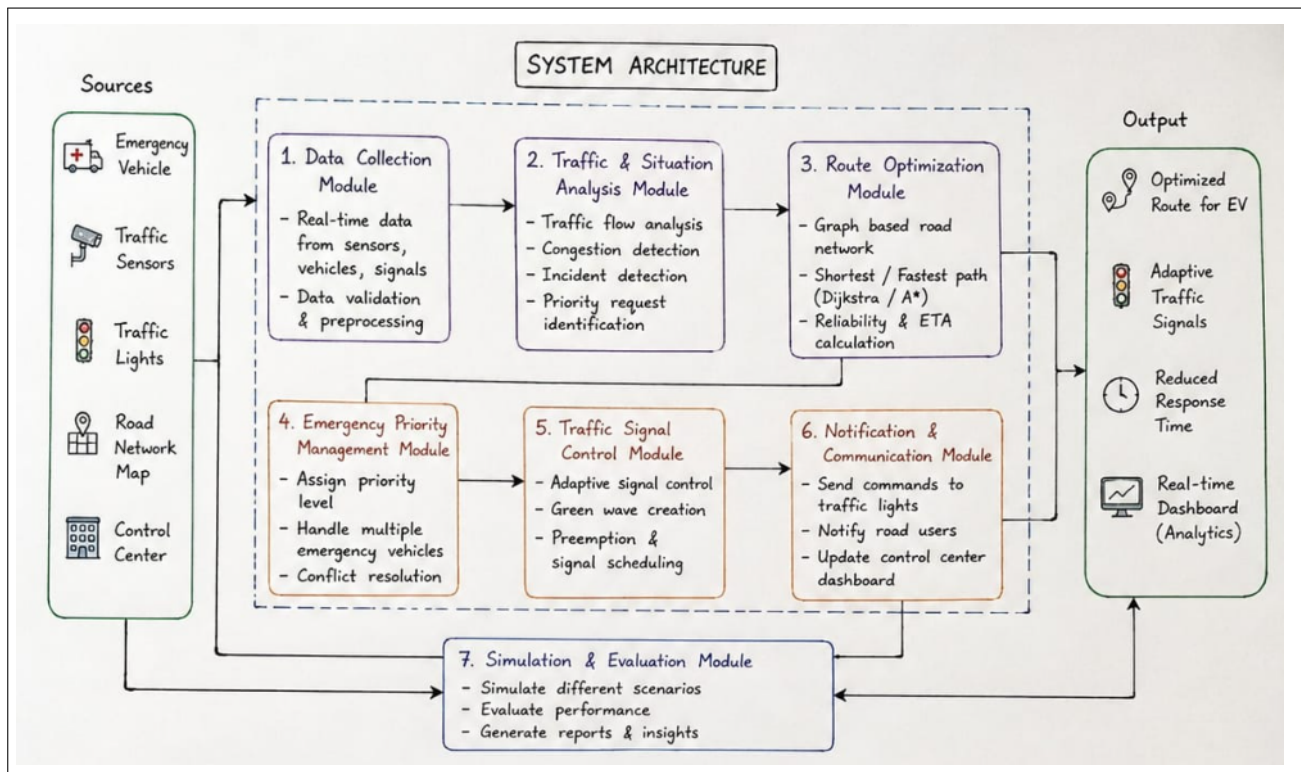
Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

We rebuilding an emergency response system in C++. The highway system is represent graph. Graphs have been used extensively in transportation systems(intersections) and provide large body of knowledge in graph theory and related areas so that established algorithms may be used in routing. The routing algorithm will search the graph from source location to a destination and find what seems to be the best route at the time. In addition traffic information can be added to each segment of road. This provides a method for simulating the impact of different traffic conditions upon the routing algorithms ability to find an acceptable route. After determining the route through the network, the simulation proceeds to determine which intersections are needed to give priority control for emergency vehicles. All traffic signal and vehicles in the simulation are simulated and there is no involvement with actual traffic signals or vehicles.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

By the end of the project, we expect to have a working C++ prototype in hand. The program should be able to model the road network, take in an emergency vehicle's starting point and destination, and work out a suitable route. Beyond that, it should also pick out which traffic signals along that route are candidates for priority.

Testing will happen in a simulated environment. We'll set up a range of road layouts and traffic conditions, then compare the standard route against the optimized one — looking mainly at response time, distance covered, and time lost waiting at intersections.

By the end, we'll have the prototype itself, along with documentation covering the system architecture and the test results. Down the line, the same approach could be extended further — incorporating live traffic data, handling multiple emergency vehicles at once, and adjusting routes dynamically as conditions change.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

For this project, we assume that the road network and basic traffic information are available in digital form. The emergency vehicle will have a known starting point and destination. We also assume that the traffic signals in our simulation can receive a priority instruction. The first version will only be a software model, so it will not need physical traffic lights, sensors or other roadside hardware. The traffic conditions used for testing will be created by us in the simulation and therefore may not cover every situation that can happen in a real city.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

- 1. Shaaban et al., "A Strategy for Emergency Vehicle Preemption and Route Selection," 2019.*
- 2. Dodia, A., Kumar, S., Rani, R., Pippal, S. K., and Meduri, P., "EVATL: A novel framework for emergency vehicle communication with adaptive traffic lights for smart cities," IET Smart Cities, vol. 5, pp. 254–268, 2023. DOI: 10.1049/smc2.12068.*
- 3. Dijkstra, E. W., "A note on two problems in connexion with graphs," Numerische Mathematik, vol. 1, pp. 269–271, 1959. DOI: 10.1007/BF01386390.*
- 4. Federal Highway Administration, "Traffic Signal Timing Manual: Chapter 9 – Advanced Signal Timing Topics," U.S. Department of Transportation.*